under anoxic conditions by phototrophic sulphur bacteria like *Chromatium* and *Chlorobium*.

Sulphate reduction

When sulphate is present in habitats, different groups of microorganisms are capable of carrying out sulphate reduction.

Beijerinck described the use of sulphate (SO_4^{2-}) as a terminal electron acceptor during anaerobic respiration to form sulphide (H_2S). This process is called **Dissimilatory Sulphate Reduction(DSR)**. The anerobic bacteria capable of carrying out DSR are *Desulfovibrio, Desulfococcus, Desulfotomaculum* (Figure 11.6) This reaction by sulphate reducers requires organic carbon sources like pyruvate or lactate. H_2S accumulated in such habitats by the action of sulphate reducers is toxic to aerobic organisms.

The reduction of sulphate to H_2S, for building up aminoacids and proteins is called as **assimilatory sulphate reduction**. The H_2S thus produced is immediately incorporated into organic compounds.

Sulphur reduction ($S^0 \rightarrow H_2S$):

The dissimilative sulphur reducing bacteria can reduce elemental sulphur to hydrogen sulphide. Example: *Desulfuromonas*, an obligate anaerobe. Under aerobic conditions, organisms like *Pseudomonas, Proteus* and *Salmonella* are also capable of performing this reaction.

Organic sulphur compounds reduction/oxidation

Organic sulphur compounds like dimethyl sulphide can be used as carbon and energy source for many microorganisms.

Desulfurylation

It is a process where organic sulphur compounds are used up by microorganisms for energy to produce H_2S.

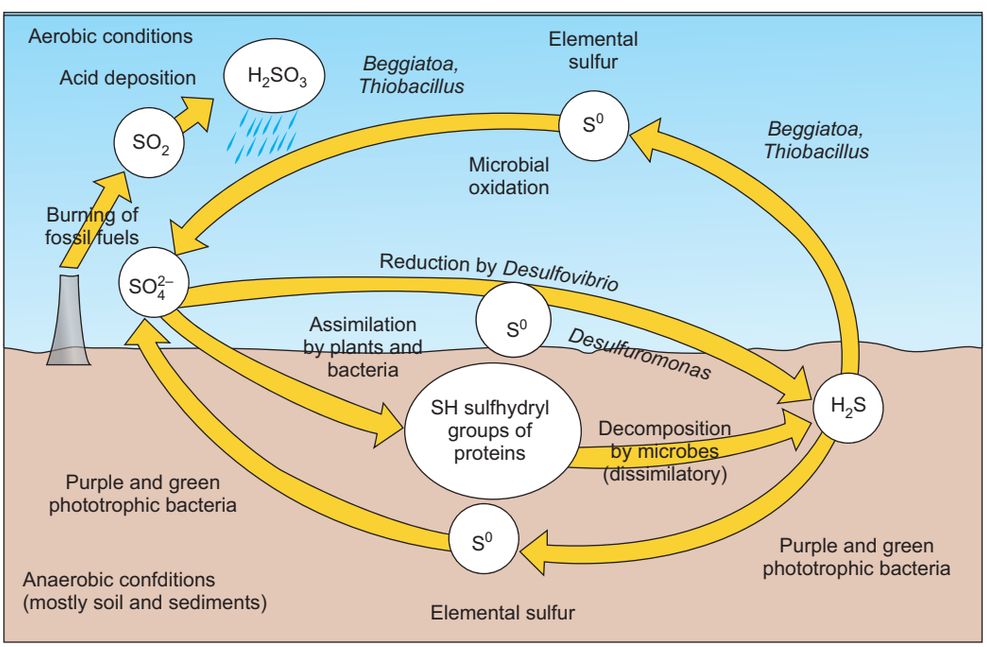


Figure 11.6: Sulphur cycle

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